

PROJECT DEVELOPMENT

5. PROJECT DEVELOPMENT PHASE

5.1 INTRODUCTION

The project is developed in the ServiceNow platform by creating users, roles, a custom table, records, and Script-Controlled Access Control Lists (ACLs).

The development process consists of user and role creation, table creation, record creation, and implementation of Read, Create, Write, and Delete ACLs. ♦

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5.2 USER AND ROLE CREATION

Procedure

Log in to ServiceNow with administrator access.

Navigate to User Administration → Users.

Click New.

Create the user EEE User.

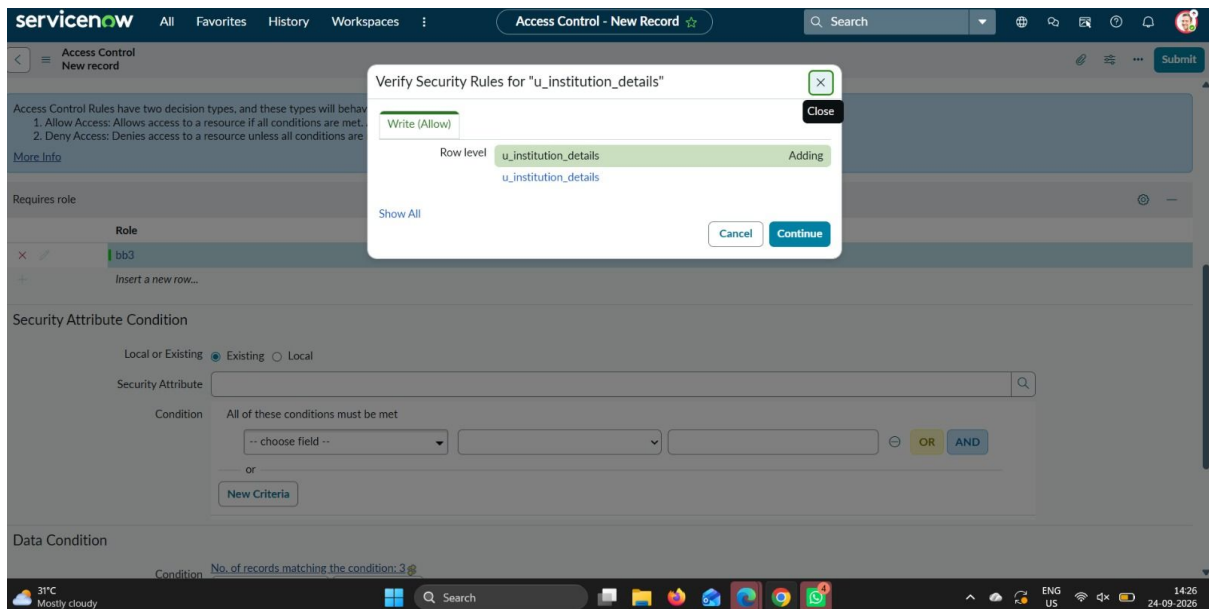
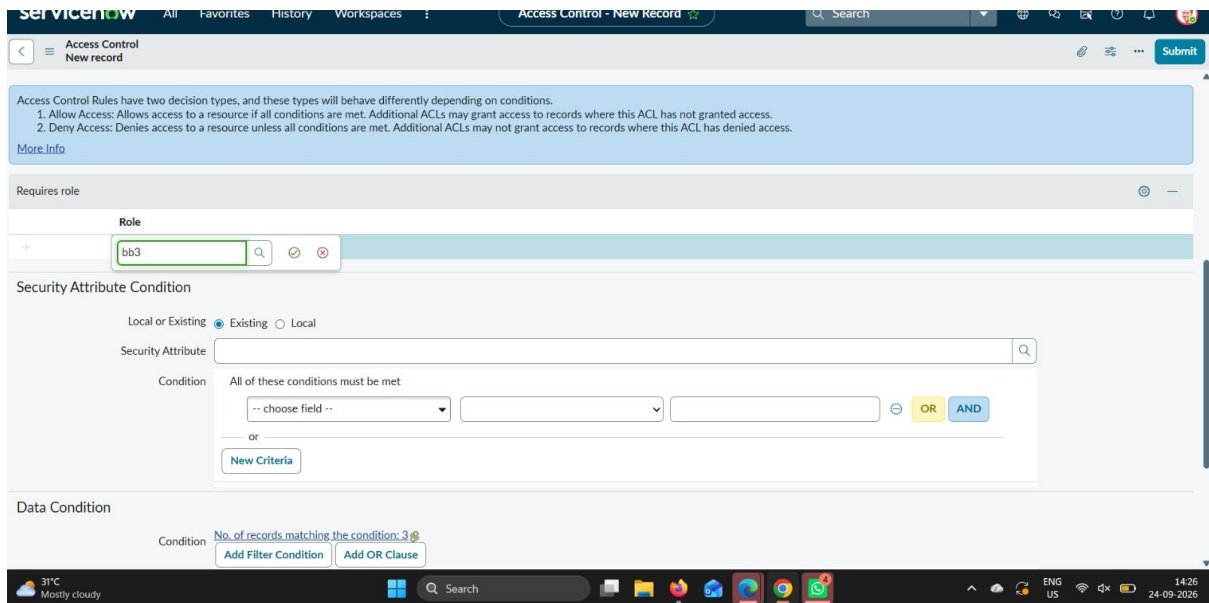
Create the roles bb1, bb2, bb3, and bb4.

Assign the required roles to the EEE User.

The screenshot displays the ServiceNow User Administration interface for creating a new user. The top navigation bar includes 'All', 'Favorites', 'History', 'Workspaces', and 'Admin'. The current user is 'User - EEE User'. The form fields are as follows:

- User ID:
- First name:
- Last name:
- Title:
- Department:
- Password needs reset: ☐
- Locked out: ☐
- Active: ☒
- Identity type:
- Internal Integration User: ☐
- Email:
- Language:
- Calendar integration:
- Time zone:
- Date format:
- Business phone:
- Mobile phone:
- Photo:

Buttons at the bottom include 'Update', 'Set Password', and 'Delete'. Below the form, there are 'Related Links' for 'View linked accounts', 'View Subscriptions', and 'Reset a password'. At the very bottom, there are tabs for 'Entitled Custom Tables', 'Roles', 'Groups', 'Delegates', 'Subscriptions', and 'User Client Certificates'.



The project instructions specify the EEE User and custom roles bb1, bb2, bb3, and bb4. ♦

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5.3 TABLE AND FIELD CREATION

Procedure

Open System Definition → Tables.

Click New.

Set the table label as Institution Details.

Set the table name as u_institution_details.

Create the required fields.

Save the table.

Required fields:

Student Roll Number – Auto Number

Student Name – Reference – User

Faculty Name – Reference – User

Branch – Choice (ECE, EEE, CSE)

Email – String

Phone Number – String

Description – Multi String

The screenshot displays the ServiceNow interface for configuring a table named 'Institution Details'. The 'Columns' tab is active, showing a list of dictionary entries. The 'Student Roll Number' and 'Student Name' fields are highlighted with green borders, indicating they are required fields. The 'Student Name' field is also linked to a user reference.

Column label	Type	Reference	Max length	Default value	Display
Updates	Integer	(empty)	40		false
Created	Date/Time	(empty)	40		false
Updated by	String	(empty)	40		false
Updated	Date/Time	(empty)	40		false
Created by	String	(empty)	40		false
Sys ID	Sys ID (GUID)	(empty)	32		false
Student Roll Number	String				false
Student Name	String				false

servicenow All Favorites History Workspaces Admin Table - Institution Details

Table Columns for text Search 1 to 6 of 6 New

Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display
Updates	Integer	(empty)	40	40	false
Created	Date/Time	(empty)	40	40	false
Updated by	String	(empty)	40	40	false
Updated	Date/Time	(empty)	40	40	false
Created by	String	(empty)	40	40	false
Sys ID	Sys ID (GUID)	(empty)	32		false
Student Roll Number	String				false
Student Name	Reference	User			false
Faculty Name	Reference	User			false
Branch	Choice	String			false

Insert a new row...

Delete Update Delete All Records

Related Links

Form Builder
Sign Form
Route Form
Route List

The table and field requirements are specified in the project instructions. ♦

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5.4 RECORD CREATION

Multiple records are created in the table using different branch values:

ECE

EEE

CSE

servicenow All Favorites History Workspaces Admin Institution Details

Institution Details Branch Search Actions on selected rows... New

Branch	Description	Faculty Name	Phone Number	Student Name	Student Roll Number
EEE		(empty)		Alejandra Prenatt	STU200
ECE		(empty)		Abel Tuter	STU100
CSE		(empty)		Amos Linnan	STU300

1 to 3 of 3

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5.5 READ ACL IMPLEMENTATION

A Read ACL is created for the u_institution_details table.

Configuration:

Type: Record

Operation: Read

Name: u_institution_details

Active: True

Advanced: True

Required Role: bb1

Data Condition: Branch is EEE

The ACL script checks the administrator and bb1 roles before allowing access. ♦

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The screenshot shows the 'Access Control - New Record' form in the ServiceNow interface. The form is titled 'Access Control - New Record' and includes a warning message: 'Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.' The form fields are as follows:

- Type:** record (dropdown)
- Operation:** read (dropdown)
- Decision Type:** Allow If (dropdown)
- Admin overrides:** ☒
- Protection policy:** -- None -- (dropdown)
- Name:** * (dropdown)
- Description:** (text area)
- Applies To:** Not a valid table name (error message), Add Filter Condition, Add OR Clause (buttons)
- Application:** Global (dropdown)
- Active:** ☒
- Advanced:** ☐

Below the form, there is a 'Conditions' section with a blue box containing the following text:

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type: record

* Operation: write

Decision Type: execute

Admin overrides: query_match, conditional_table_query_range

Protection policy: query_range

* Name:

Description: write

Applies To:

Application: Global

Active: ☒

Advanced: ☐

Conditions

Access Control Rules have two decision types: Allow and Deny. The Allow decision type is used to grant access to records based on conditions. The Deny decision type is used to deny access to records based on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.

2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

More Info

5.6 CREATE ACL IMPLEMENTATION

A Create ACL is created for the u_institution_details table.

Configuration:

Type: Record

Operation: Create

Name: u_institution_details

Active: True

Required Role: bb2

No data condition is required for the Create ACL. ♦

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Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type: record

* Operation: write

Decision Type: Allow If

Admin overrides: ☒

Protection policy: -- None --

* Name: Institution Details [u_institution_details]

Description: View results in new tab

Applies To: No. of records matching the condition: 3

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.

2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

5.7 WRITE ACL IMPLEMENTATION

A Write ACL is created for the u_institution_details table.

Configuration:

Type: Record

Operation: Write

Name: u_institution_details

Active: True

Required Role: bb3

No data condition is required for the Write ACL. ♦

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5.8 DELETE ACL IMPLEMENTATION

A Delete ACL is created for the u_institution_details table.

Configuration:

Type: Record

Operation: Delete

Name: u_institution_details

Active: True

Required Role: bb4

No data condition is required for the Delete ACL. ♦

Script-Controlled ACL-Micro Projects.pdf

5.9 DEVELOPMENT RESULT

The ServiceNow project was developed successfully by implementing user roles, a custom table, student records, and ACLs for Read, Create, Write, and Delete operations.

The implemented ACLs provide record-level access control based on roles and the configured branch condition. ♦